GridPulse: Forecasting Tomorrow's Electricity Demand

Ridge Regression and Temporal Convolutions

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1 Abstract

GridPulse forecasts the next 24 hourly values of Great Britain's National Demand using the preceding 168 hours and deterministic calendar features. The project compares a weekly seasonal baseline, multi-output Ridge regression and a causal temporal convolutional network (TCN). Three expanding-window validation blocks select the model family before evaluation on a later contiguous holdout.

The recorded benchmark contains 102,863 hourly observations from January 2015 to September 2026. It produces 4,278 daily forecast windows: 3,422 for development and 856 for the final holdout, corresponding to 20,544 hourly predictions. Ridge achieves mean validation MAE of 982 MW, compared with 1,135 MW for the TCN and 2,050 MW for the weekly baseline. Ridge is selected in the supplied experiment and the losing neural model is not evaluated on the final holdout.

The saved holdout MAE is 1,164 MW for Ridge and 2,053 MW for the weekly baseline, a 43.29% reduction. RMSE is 1,661 MW versus 2,781 MW. On the highest-demand decile, MAE is 1,273 MW versus 2,893 MW. A pooled residual interval with nominal 80% coverage covers only 74.96% of holdout hours.

The comparison supports this Ridge specification on the studied retrospective task. It does not establish a general advantage of linear models over neural networks: Ridge receives an explicit future-calendar matrix and a full week of history, while the implemented TCN's effective receptive field is at most 61 hours.

The report audits the original results and a full replay using freshly downloaded history truncated to the recorded endpoint. It distinguishes numerical reproducibility from real-time validity: the source is revised and published 21 days in arrears. A timely input vintage and improved interval calibration are prerequisites to an operational trial. Nine repository tests now pass, including two new gap-handling regressions. The companion evidence contains the processed replay series, forecast ledger and verification details.

2 Introduction

Electricity-demand forecasts can help a planning process anticipate future load patterns. A useful modelling question is whether recent demand and calendar structure alone can predict the next day's hourly profile. This project addresses that statistical question; it does not simulate how a dispatcher or trader would act on a forecast.

Daily, weekly and annual cycles make seasonal copying a meaningful baseline. A flexible model should therefore be evaluated against the corresponding hour one week earlier, rather than against an implausible constant prediction. Historical recurrence is strong, but changes in level and shape can make literal weekly copying inaccurate.

The first challenger is Ridge regression. It combines the full historical window and known calendar features in a direct multi-output linear forecast. A quadratic coefficient penalty controls instability among correlated lags. The neural challenger uses dilated causal convolutions and residual blocks to learn nonlinear temporal patterns.

The central empirical question is which supplied specification performs better on later observations after development-only comparison. A second question is whether uncertainty estimates remain reliable after the final refit. The experiment evaluates both the average error and behaviour at high realised demand levels.

The project demonstrates data preparation, chronological model selection, forecasting and a local API example. The stronger contribution is the complete comparison and its limitations. In particular, the available inputs differ between the linear and neural specifications, and the historical target is not an archived real-time information stream. Both distinctions matter when interpreting a large baseline improvement.

3 Data, Target and Time Conventions

3.1 National Demand and the recorded data window

The source is NESO's Historic Demand Data, retaining National Demand (ND) actuals. ND is an operational generation-requirement measure. It excludes station load, hydro pumping and interconnector exports; it is not identical to total end-use consumption [3]. Values are measured in MW. They are power levels, not daily MWh totals.

The recorded hourly series runs from 1 January 2015 at 00:00 UTC to 25 September 2026 at 22:00 UTC. It contains 102,863 hours with no missing values remaining after processing, no negative hourly values and no duplicate hourly timestamps. Median ND is 27,472 MW. These summaries describe the processed series; they do not by themselves prove that the raw files were complete or duplicate-free.

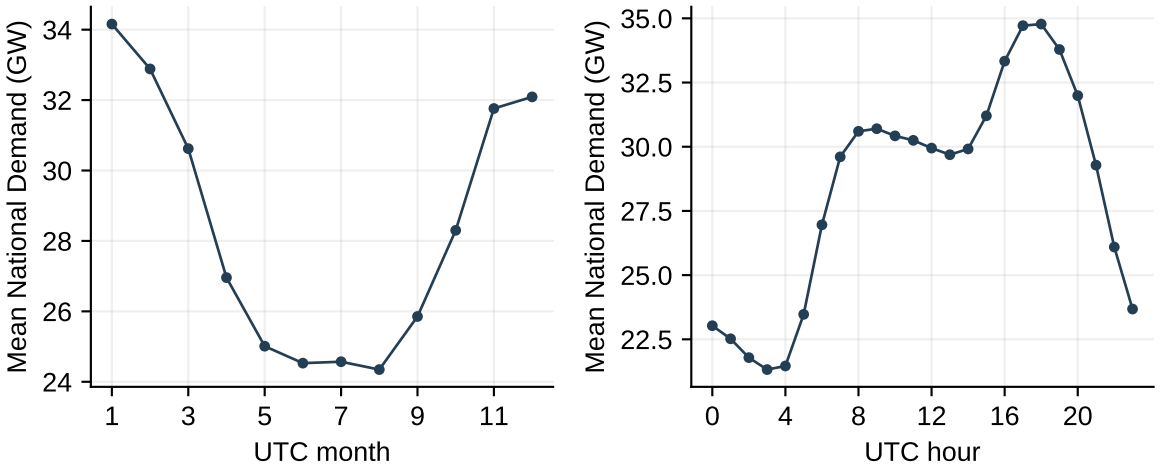


Figure 1: Descriptive monthly and hourly means from the recorded benchmark profile. These use the complete retrospective series and illustrate recurring structure; they are not an additional training feature estimated from the holdout.

3.2 Settlement time and preprocessing

Annual half-hourly CSVs use UK settlement dates and periods. The loader localises each date's midnight to Europe/London, converts it to UTC and adds the elapsed half-hours associated with the settlement period. This handles 46-period spring days and 50-period autumn days. It retains actual-labelled rows where the indicator is present; older outturn-only files lack that column.

Duplicate half-hour timestamps retain the last record. Half-hourly MW values are averaged into hourly MW values and reindexed onto a continuous UTC grid. The code time-interpolates up to two successive missing values using surrounding observations. Longer gaps can remain missing and any supervised window still containing a nonfinite value is excluded.

3.3 What information is assumed available?

The benchmark assumes that the latest hourly actuals can be used at the forecast origin. NESO describes the historical product as populated 21 days in arrears and subject to retrospective corrections [2]. Chronological splitting therefore does not establish that the same input values were published and available at the corresponding historical decision times.

This is a retrospective prediction experiment on revised actuals. An operational backtest would need timestamped data vintages or another archived timely input stream, together with the actual publication and processing delays. The study excludes future realised weather and generation, but that exclusion alone does not resolve the demand-vintage issue.

The stored origin timestamp labels the last input hour. An hourly mean is fully observed only after that hour completes. A live implementation would need to define release time relative to completed measurements and the first target interval, rather than assuming an instant reading at the start of the labelled hour.

3.4 Supervised forecast windows

For hourly demand L_t, a window ending at t contains

$$\mathbf{x}_t = (L_{t-167}, \ldots, L_t), \qquad \mathbf{y}_t = (L_{t+1}, \ldots, L_{t+24}). \tag{1}$$

Origins advance by 24 hours, so forecast targets do not overlap. In the recorded run, target blocks start at 00:00 UTC, beginning on 8 January 2015. This is a fixed 24-hour UTC task; it is not a variable-length UK settlement-day forecast at clock changes.

For each timestamp, seven deterministic features encode sine and cosine pairs for UTC hour, weekday and day of year, plus a weekend indicator. Future calendar values are known at prediction time. No holiday indicator, weather forecast, outage information or demand-forecast product is supplied to the models.

The total is 4,278 windows. The final target block starts on 24 September 2026 and ends at 23:00 UTC that day. The saved `test_end` field is the last target-block start, rather than the final predicted hour. The unused trailing partial day is not converted into a shorter forecast.

4 Forecasting Models

4.1 Weekly seasonal baseline

For forecast step $h = 1, \ldots, 24$, the baseline is

$$\widehat{L}_{t+h} = L_{t+h-168}. \tag{2}$$

It copies the corresponding hour from the previous week. The helper also supports a prior-day baseline, but the saved controlled comparison and headline improvement use the weekly version. No unreported prior-day holdout result is inferred.

4.2 Multi-output Ridge regression

The Ridge feature vector flattens all 168 historical hours, each with one load and seven calendar channels, and appends the next 24 hours' calendar matrix. This gives $168 \times 8 + 24 \times 7 = 1,512$ input features. The model emits all 24 outputs directly.

Within a fitting block, load and target values are normalised using the training-history mean and standard deviation. The Ridge pipeline also fits a feature-wise standard scaler on that block. Calendar features and all horizon outputs enter a multi-output linear fit:

$$\min_{B,\mathbf{b}} \ \|Y - ZB - \mathbf{1}\mathbf{b}^{\mathsf{T}}\|_F^2 + \alpha\|B\|_F^2, \qquad \alpha = 10. \tag{3}$$

Here Z is the scaled feature matrix and the intercept is unpenalised. The penalty strength is fixed in the supplied specification; the experiment selects among model families rather than reporting a separate alpha-tuning study.

Direct prediction avoids recursively feeding one predicted hour into the next. It does not remove dependence between hourly errors, nor guarantee stable performance when demand patterns change. The fitted coefficients combine correlated historical and calendar inputs, so they should not be interpreted as isolated causal effects.

4.3 Causal temporal convolutional network

The TCN receives the historical eight-channel window. Each of four residual blocks has two left-padded convolutions, ReLU activations and dropout. Dilation rates are 1, 2, 4 and 8, with kernel size three, 32 hidden channels and dropout 0.1. The final hidden position feeds a linear 24-output head.

4.4 Neural training and effective context

A causal convolution has the form

$$z'_t = \sum_{k=0}^{K-1} W_k z_{t-dk}, \tag{4}$$

so its within-window computation does not access later positions. With two convolutions per block, the maximum receptive field of this architecture is

$$1 + 2(K-1) \sum_{d \in \{1,2,4,8\}} d = 1 + 2 \cdot 2 \cdot 15 = 61 \text{ hours.} \tag{5}$$

Although 168 hours are supplied, the final output cannot depend on the oldest 107 positions. A direct perturbation check confirms that changing those positions leaves this architecture's evaluation-mode output unchanged.

Training uses Huber loss in normalised target units, AdamW at learning rate 10^{-3}, batch size 128, gradient-norm clipping at one and at most 30 epochs. A trailing 10% of each outer training block supplies early stopping, with patience five. The network returns its best saved stopping state. Seeds vary by fold from 42.

Table 1: Information and fitting differences between the supplied challengers.

Property	Ridge	TCN
Maximum accessible load history	168 hours	61 hours
Explicit future-calendar matrix	Yes	No
Outer training block used for coefficient/weight fitting	Full block	First 90%
Primary fitting loss	Squared error	Huber loss
Family selection criterion	Validation MAE	Validation MAE

The TCN has historical calendar channels, but the explicit future-calendar matrix passed to Ridge is not passed into its model. Consequently, the experiment compares these complete specifications. It does not isolate model class under equal information, equal context and equal fitting budget. The observed Ridge advantage is useful for selection, while a general claim that neural forecasting is inferior would exceed the design.

5 Chronological Evaluation and Error Measures

5.1 Development folds and final holdout

The first 80% of forecast windows form development data: 3,422 origins. The final 856 form the contiguous holdout. The trainer sets each validation block to 10% of development, rounded down to 342 origins. The configuration's `validation_days: 60` field is not used to set these actual blocks.

Table 2: Actual outer validation blocks. Dates identify 24-hour target-block starts.

Fold	First target date	Last target date	Origins
1	2021-07-31	2022-07-07	342
2	2022-07-08	2023-06-14	342
3	2023-06-15	2024-05-21	342

Training expands from 2,396 to 2,738 to 3,080 origins. Scaling is fitted within each outer training block. TCN stopping uses a trailing sub-block of that training period; it does not use outer validation or holdout outcomes. The outer-block load normaliser is shared across that internal split.

The family with the lowest average validation MAE is refitted for the holdout. Here it is Ridge, fitted to all 3,422 development origins. It remains fixed during evaluation; the history updates as new actual demand observations become available in the retrospective series. The final target blocks run from 22 May 2024 to 24 September 2026, ending at 23:00 UTC on the latter date. This is a single examined holdout, rather than evidence of prospective deployment.

5.2 Metrics and high-demand errors

MAE averages $|\widehat{L} - L|$ over all 20,544 target hours. RMSE is the square root of mean squared error. Symmetric MAPE averages $200|\widehat{L}-L|/(|L|+|\widehat{L}|)$. MASE divides MAE by the development sample's weekly-naive absolute-error scale; it does not divide by the holdout baseline's MAE.

The source's peak-hour MAE selects target observations at or above the evaluation sample's 90th demand percentile. It is a retrospective high-demand subgroup, not a predeclared 16:00–19:00 clock-hour window or a metric on predicted peaks. That distinction is retained throughout the report.

6 Validation and Holdout Results

6.1 Model selection and main errors

Table 3: Recorded validation MAE in MW. Ridge wins each supplied fold.

Fold	Train	Validation	Ridge	TCN	Weekly
1	2,396	342	1,009.3	1,182.5	1,985.3
2	2,738	342	962.5	1,106.6	2,079.2
3	3,080	342	973.9	1,116.6	2,086.0
Mean	–	–	981.9	1,135.2	2,050.1

Ridge's mean validation MAE is 13.5% lower than the TCN's. The ranking favours the simpler supplied specification consistently across the three blocks. This comparison uses a single architecture and fixed Ridge penalty, not an exhaustive search over either model family.

Table 4: Recorded final holdout, 856 daily forecasts.

Model	MAE (MW)	RMSE (MW)	sMAPE	MASE
Ridge	1,164.4	1,661.2	4.637%	0.587
Weekly baseline	2,053.3	2,781.3	8.081%	1.035

Ridge lowers holdout MAE by 43.29% and RMSE by 40.27% relative to the weekly baseline. These are reductions in error, rather than percentage increases in forecasting accuracy or financial returns. The TCN is not scored on the final holdout after losing selection; no test score for it is supplied or invented.

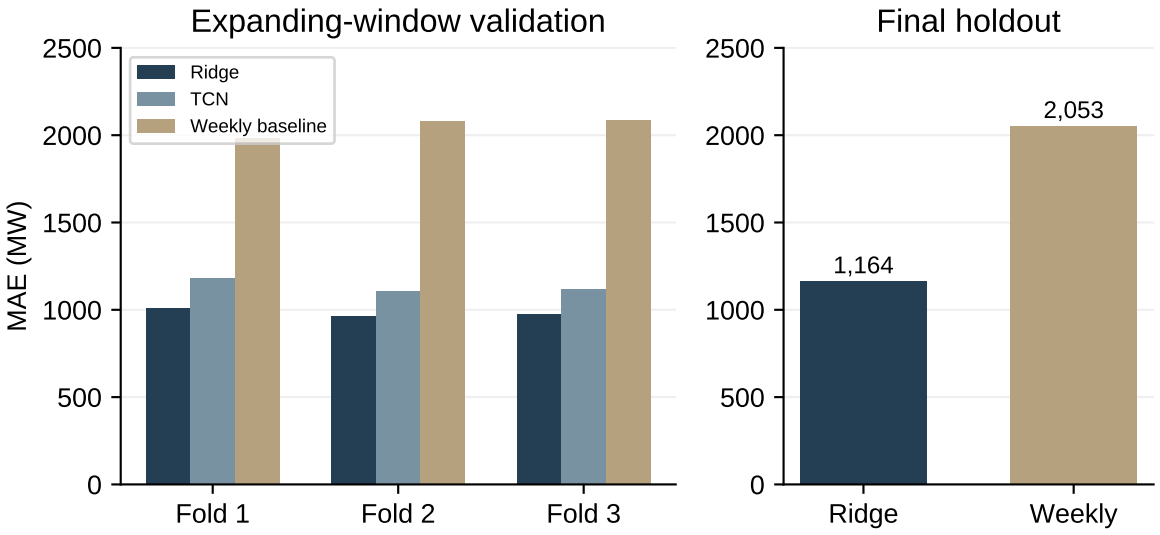


Figure 2: Recorded validation and holdout MAE. The TCN appears only in the development comparison. The final evaluation follows the selected Ridge specification.

6.2 High-demand observations and temporal slices

On the highest-demand decile, recorded Ridge MAE is 1,272.6 MW, versus 2,892.9 MW for weekly copying, a 56.01% reduction. This demonstrates smaller point errors on the selected high-demand observations in this retrospective sample. It does not measure reserve adequacy, shortage risk or the cost of an underforecast.

The fresh-download replay exports every holdout prediction. The following fixed-holdout slices summarise those forecasts by target year. They were computed for this report after model fitting and were not used to choose a family or retune a model.

Table 5: Replay holdout slices by UTC target year. First and last years are partial periods.

Year	Hours	Ridge MAE	Weekly MAE	Reduction
2024	5,376	1,021.6	1,787.2	42.8%
2025	8,760	1,150.7	2,180.7	47.2%
2026	6,408	1,303.0	2,102.4	38.0%

Ridge MAE rises from 1,021.6 MW to 1,303.0 MW across these evaluated segments, while remaining below the baseline in each. Their seasonal compositions differ, so this is a descriptive comparison rather than an isolated test of structural change.

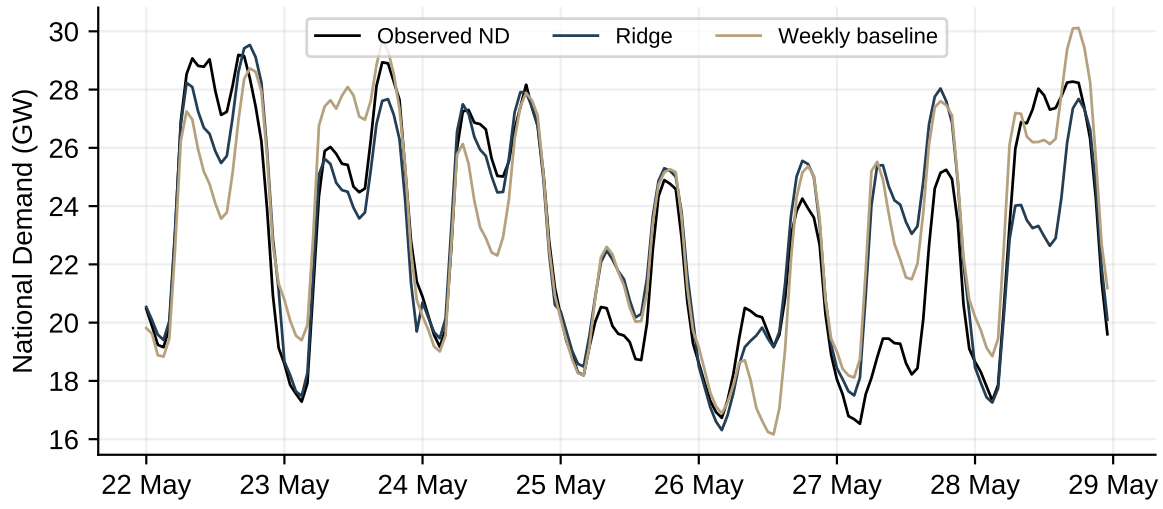


Figure 3: The first seven holdout target blocks, fixed by date order. Observed demand is compared with Ridge and weekly forecasts. The example illustrates hourly profile behaviour; the full holdout determines the aggregate result.

7 Prediction Intervals and Hourly Diagnostics

7.1 Pooled residual intervals

The selected family's outer-validation residuals are pooled over three blocks and all forecast hours. With $e = L - \widehat{L}$ and empirical residual quantiles $q_{0.1}$ and $q_{0.9}$, the reported interval is

$$[\widehat{L} + q_{0.1},\ \widehat{L} + q_{0.9}]. \tag{6}$$

The recorded offsets are $-1,677.8$ MW and $+1,580.3$ MW, giving a constant width of 3,258.1 MW. A pooled residual band is straightforward to inspect, but it does not adapt to target hour, season or forecast demand level. It is not a calibrated conformal or conditional-coverage guarantee.

Nominal 80% coverage falls to 74.96% on the final holdout: undercoverage of 5.04 percentage points. Mean validation and holdout error also differ. A fixed residual distribution from development need not remain suitable after a final refit and a later temporal shift.

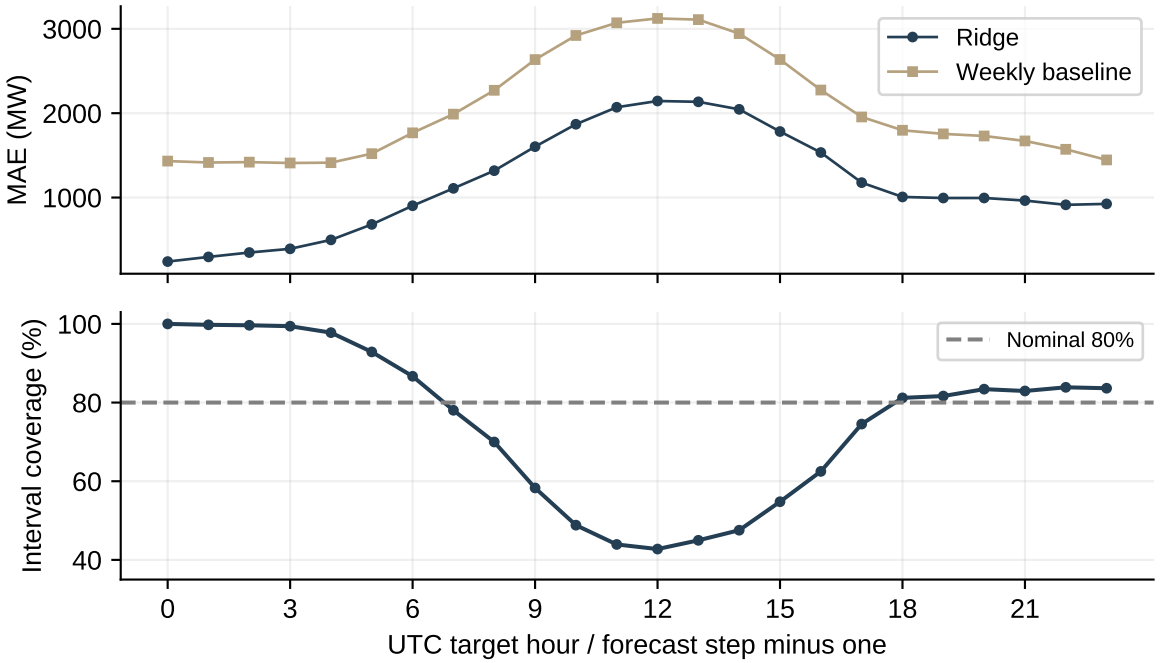


Figure 4: Supplementary replay diagnostics by UTC target hour. Because every target block begins at midnight, clock hour is also forecast step minus one; the two effects cannot be separated here. Upper panel: MAE. Lower panel: empirical interval coverage against its nominal target.

In the replay, hourly coverage ranges from 42.8% to 100%. Overall coverage by target year falls from 79.3% in the evaluated 2024 segment to 74.7% in 2025 and 71.7% in the evaluated 2026 segment. These retrospective slices identify a practical weakness in a single pooled band; they do not isolate whether horizon, season or changing demand conditions cause it.

The 24 hourly errors within a forecast day are dependent, and errors can remain dependent across days. The report provides point comparisons rather than treating 20,544 hours as independent evidence for a significance claim. A future interval study should evaluate dependence, conditional coverage and adaptation on a separate later period.

8 Verification, Limitations and Next Milestone

8.1 What this replay establishes

The earlier full replay passed all seven then-existing repository tests and 38 independent evidence checks. The revised code passes nine tests. The public annual files were downloaded again, processed by the supplied loader and truncated to the recorded endpoint. The processed quality summaries match the saved run. The original download manifest and raw vintage are absent, so byte identity with the original files is not established. The new source identifiers and hashes are retained separately.

The complete supplied trainer was run with the recorded model settings: three expanding folds, neural stopping, family selection and final evaluation. The replay again selects Ridge. Its holdout MAE is 1,164.4299 MW, versus 1,164.4194 MW recorded, a difference of 0.0105 MW. Mean TCN validation MAE is 1,136.85 MW, versus 1,135.23 MW recorded. These small numerical differences preserve the reported ranking; their causes cannot be isolated without the original raw vintage and runtime. Exact comparisons are retained in the verification file. Every holdout target and weekly prediction was rebuilt from the processed series; aggregate errors and interval coverage were recomputed from the new prediction ledger. The verification JSON records exact comparisons and scope.

A supplementary check confirms the TCN context limit and both UK clock transitions. The replay has no missing hours before interpolation. The code review fixes partial filling of longer gaps: only complete internal runs of at most two hours are now filled. Two regression tests cover this rule. Unused configuration defaults were removed. Neither change affects the complete replay series. Short-gap interpolation still uses a later endpoint, so it remains retrospective; future availability tests must cover preprocessing as well as train/test indices.

8.2 Limits on the interpretation

The strongest inference concerns this input series, these model specifications and this historical split. Real-time publication delays and revisions remain untested. The comparison omits archived weather forecasts, holiday detail and a competitive equal-context neural search. Ridge and TCN do not receive identical information or identical effective fitting samples.

The supplied benchmark is already examined; new hourly and yearly slices are retrospective diagnostics. There are no independently validated dispatch outcomes, economic savings or trading returns. The local API and container files are demonstrations of packaging and inference, not evidence of live uptime, security hardening or operational forecast performance.

8.3 A realistic next milestone

Freeze a new period and define release times using timely archived load vintages. Make gap filling causal. Compare Ridge with a full-week TCN receiving the same future calendar information and fitting windows. Use inner chronological selection, then evaluate daily errors, peak underforecasts and conditional interval coverage. Retain a dated prediction ledger and input-vintage identifiers. This is proposed work.

9 Conclusion

GridPulse converts a public operational series into a direct next-24-hour forecasting task, with settlement-time handling, expanding validation and a later contiguous evaluation block. The supplied Ridge model has lower validation error than the TCN in each fold, and its recorded holdout MAE is 43.29% below weekly copying. It also reduces errors on the highest-demand decile.

The evidence supports using this simpler specification for the studied retrospective benchmark. It does not establish that linear modelling is intrinsically better, because the challengers differ in accessible history, calendar inputs and fitting samples. The under-covered residual intervals and unmodelled publication delays also limit operational interpretation.

The next step is a fair comparison under equal information and a verified release-time boundary, followed by improved uncertainty assessment. The present project supplies the implementation, benchmark, replay evidence and clearly identified gaps needed to make that next test concrete.

10 Reproducibility and Sources

This report uses repository commit `be8e0c15049d` [1]. The companion verification JSON records the execution environment, existing tests, independent checks, new input hashes and exact differences from the saved benchmark. The evidence archive contains the processed replay series, new source metadata, fitted Ridge bundle, full holdout ledger and supplementary diagnostics. A small wrapper reruns the pinned trainer on the included processed series.

The newly downloaded source was clipped to 25 September 2026 at 22:00 UTC; data beyond that point were excluded from fitting and reporting. No new test outcome was used to tune the recorded specification. The original and new source-manifest hashes are kept distinct. A matching model score is a numerical reproduction, rather than proof of the original raw data vintage or historical publication availability.

The main entry points are the versioned data loader, model implementations and training CLI. The report and LinkedIn entry summarise the research; the replay archive supplies the new evidence rather than changing the published repository. Project dates and an institutional association have not been inferred from report assembly or data dates.

October 2026 review: the public research decision note records the leading comparison, evidence limits and next experiment. The pinned source versions below identify the experiments; the review changes presentation and the explicitly described code checks.

References

[1] Hussain, F. H. GridPulse. Versioned implementation, benchmark metrics and methodology. October 2026 portfolio version.

[2] National Energy System Operator. Historic Demand Data. Official source and publication/revision notes. Checked October 2026.

[3] National Energy System Operator. Historic Demand Data 2026. Official field definitions and data access. Checked October 2026.